



Niyam Bhushan

Is Your TV Watching You?

Even if it runs Linux inside?

Something's happened to my TV—it's turned intelligent. The dumbest TV shows still stream mindlessly on it, 24x7, through the cable-box, but the TV has new smarts. For starters, it's got USB and card-reader ports, so I can plug in memory cards from mobile phones and digital cameras, as well as USB thumb drives. I can then view photos and videos on its dazzlingly magnificent screen. I can also browse and play music, using the TV to view and navigate through folders, and using the built-in or connected speakers to listen to the audio. It's like a super-gigantic iPod-on-the-wall.

Stream of media consciousness

How about an out-of-the-cable-box experience? With built-in Wi-Fi, the TV can connect, sans wires, to laptops, smart-phones and routers across my home, and stream music or video from anywhere—even jumping out to the Web to stream videos from YouTube and other sites. All I need to do is aimlessly lounge on the couch, and click the remote control.

That is, I can do this after I've racked my brains to set up what is called a Universal Plug and Play (UPnP) server on my Ubuntu laptop, with something like MediaTomb, for instance. And after I've gotten configurations to work on all the Digital Living Network Alliance (DLNA) devices and gadgets across my home, to 'beam up Scotty' from anywhere. Rolling out my own Wi-Fi home-streaming is not just free air, but freedom from the hot air of proprietary software vendors as well, if you know who I mean.

Somewhere along this not-so-dull reverie, it suddenly hit me: my TV is no longer the analogue bumptin from circuit city. It's turned digital! Look at it again: it's got a computer inside, complete with CPU, RAM, storage, software, ports, network interfaces, and some real smarty-pants technology—to make the colours and picture-quality sizzle and burn your neighbours.

Frantically, I rush to do some research on the Web. Sure enough, most of these computers-masquerading-as-TVs run Linux inside. But that's just the kernel, and maybe a few other software components. Surrounding the Linux kernel is a whole ecosystem of opaque proprietary software. It's the same story with similar TV sets from Samsung, Philips, Sony, LG, and a few other leading brands.

Are you channelling me?

Pause! Is the TV watching me? It can silently 'call home' and report all my viewing and listening habits—not just via the

cable box, but via every networking device connected directly, or over Wi-Fi. The Internet 'information superhighway' is two-way, remember? This is Orwell's 1984 over Wi-Fi's 802.11. The invasion of privacy on such a massive scale, whether by vendors or by cyber-criminals who may steal your data files as well, will certainly be devastating.

In fact, the controversies and issues that digital TV will inevitably give rise to may dwarf the prime-time heckles and hoots that existing TV programming so desperately thrives on. And this time, you cannot be just a passive watcher sitting at home—the real battle's happening in your living rooms and bedrooms, as you wrestle over your digital and moral rights. A whole new set of cyber laws may have to be drafted, with certifications and legislation framed for TV vendors—to not just declare, but also provide verification of the privacy and digital-security claims they make about their devices. Hint:

We're going to all need the wisdom of Free and Open Source Software real soon now!

Until then, there is no way to find out what's happening behind the glass-screen, thanks to all the hush-hush proprietary software embedded within the TV. And this

"This is Orwell's 1984 over Wi-Fi's 802.11"

is despite it running the Linux kernel and a few *mukt* software components. Can a Linux hacker hurl a sledge hammer at Big Brother's power-point screen-cast?

SamyGO (nice name!), for example, is a buzzing hacker-project on sourceforge.net, where intrepid souls try hacking into certain Samsung TV sets to take back control. Intrepid, because how do you explain to your better half that you just permanently bricked a very expensive TV, while trying to make it run Doom? Dinner will soon become a bigger luxury than TV-dinners, for the failed hacker. Still, a digital TV that works transparently allows you freedom from vendor locks or restrictions, and gives you back control of your data, which is half the battle. The battle of the sexes, though, will be won by whoever gains possession of the remote control. 

About Niyam

Inspired by the vision of Osho. Copyright August 2010: Niyam Bhushan. freedomyugs at gmail dotcom. First published in LINUX For You magazine. Verbatim copying, publishing and distribution of this article is encouraged in any language and medium, so long as this copyright notice is preserved. In Hindi, 'mukt' means 'free-of-cost', and 'mukt' means 'with freedom.'